



IP Core Protection of Image Processing Filters With Multi-Level Encryption and Covert Steganographic Security Constraints

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Introduction

Image processing filters

- ▶ Image processing filters are mainly used to suppress either the high frequencies in the image, i.e. smoothing the image, or the low frequencies, i.e. enhancing or detecting edges in the image.
- ▶ The main objective of image processing is to extract some useful information from an image.
- ▶ From detection and recognition of license plates of vehicles on tolls (character recognition), advanced medical imagery (image analysis), biometric fingerprinting, robotics vision, and military operations to car driving automation, image processing plays a crucial role everywhere.
- ▶ Due to globalization of design supply chain, the design process of these image processing filters as a dedicated intellectual property (IP) core involves various hardware threats [1], [2].

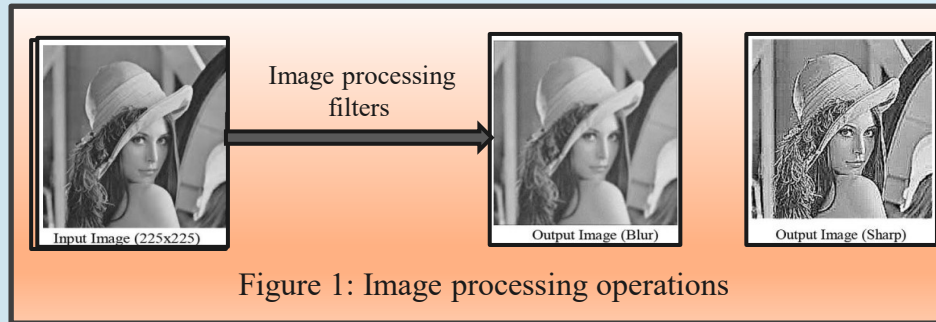
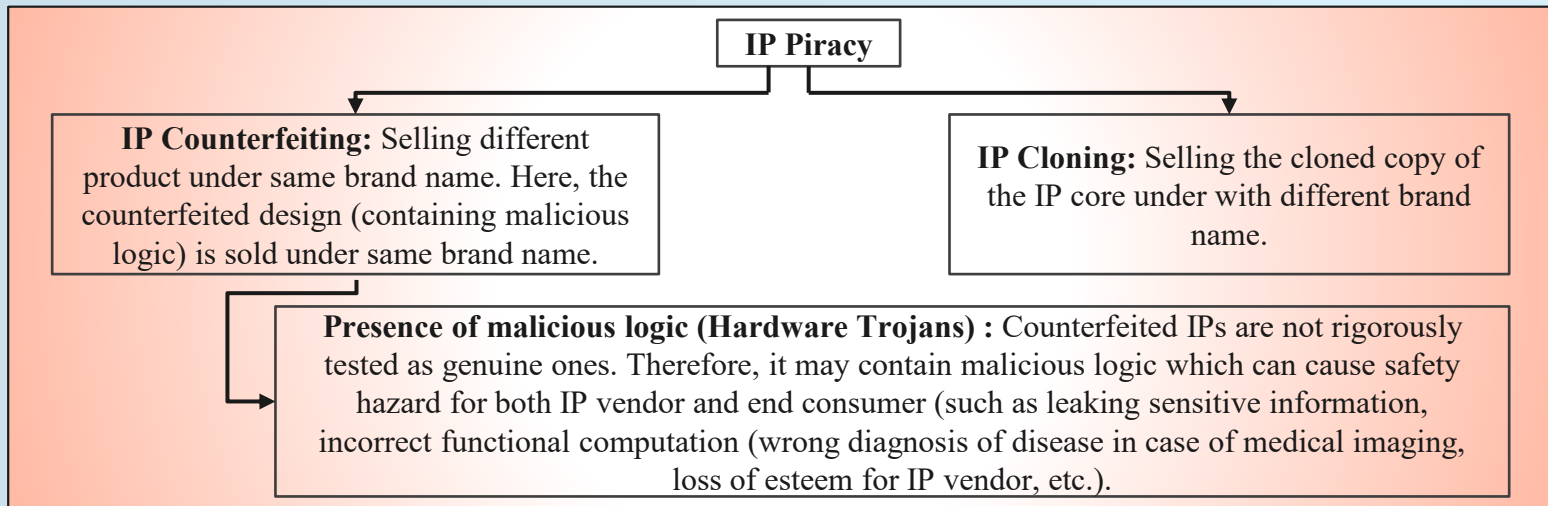


Figure 1: Image processing operations

Introduction : Hardware Threats

Security Issues associated with image processing filter IP Cores [3], [4]



Fraudulent claim of IP ownership: An adversary tries to fraudulently claim the ownership of the IP.

Therefore, it is essential to secure these image processing filter IP cores from these hardware threats.



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Previous works

Related Work



Sr. No.	Existing Work	Technique Used	Remarks
1.	D. Tsiktsiris, D. Ziouzos, and M.Dasygenis [5] (2018)	Authors discusses about the implementation of FPGA based image processing accelerators.	Does not focus on the security aspects of image processing filter hardware IPs.
2.	C. Shu, W. Pang, H. Liu, and S. Lu [6] (2019)	The paper focuses on designing of hardware accelerators for performing convolutional neural network (CNN).	Does not provide a framework for designing image processing filter IP cores using high level synthesis. Further, it also does not includes the security aspects of image processing filter hardware IPs.
3.	F. Koushanfar, I. Hong, and M. Potkonjak [7] (2005)	Hardware watermarking using two-variable (0, 1) signature encoding process.	Weak watermarking mechanism due to involvement of only two variable signature encoding process. Not robust and future proof. The watermark (original signature) inserted by watermarking technique becomes vulnerable if relevant information (like signature size, digit encoding, and digit combination) gets leaked.



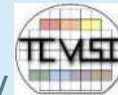
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Proposed Work

- ▶ The proposed approach based on multi-level encryption technique used for securing image processing filters IP cores. This paper discusses signature-based hardware security methodology on image processing IP cores for the first time.
- ▶ The proposed approach uses the register allocation table of the image processing application to generate secret data, which is used for multi-level encryption to determine hardware security constraints .
- ▶ The generated hardware security constraints then embedded in the image processing filter IP Cores design to authenticate genuine IP Maker.
- ▶ The huge variation in the key selection at different levels of encryption and the use of secret steganographic data for signature generation increases the robustness of the proposed hardware security methodology for image filters.



Proposed Work : Flow Diagram

Detailed flow diagram of the proposed approach

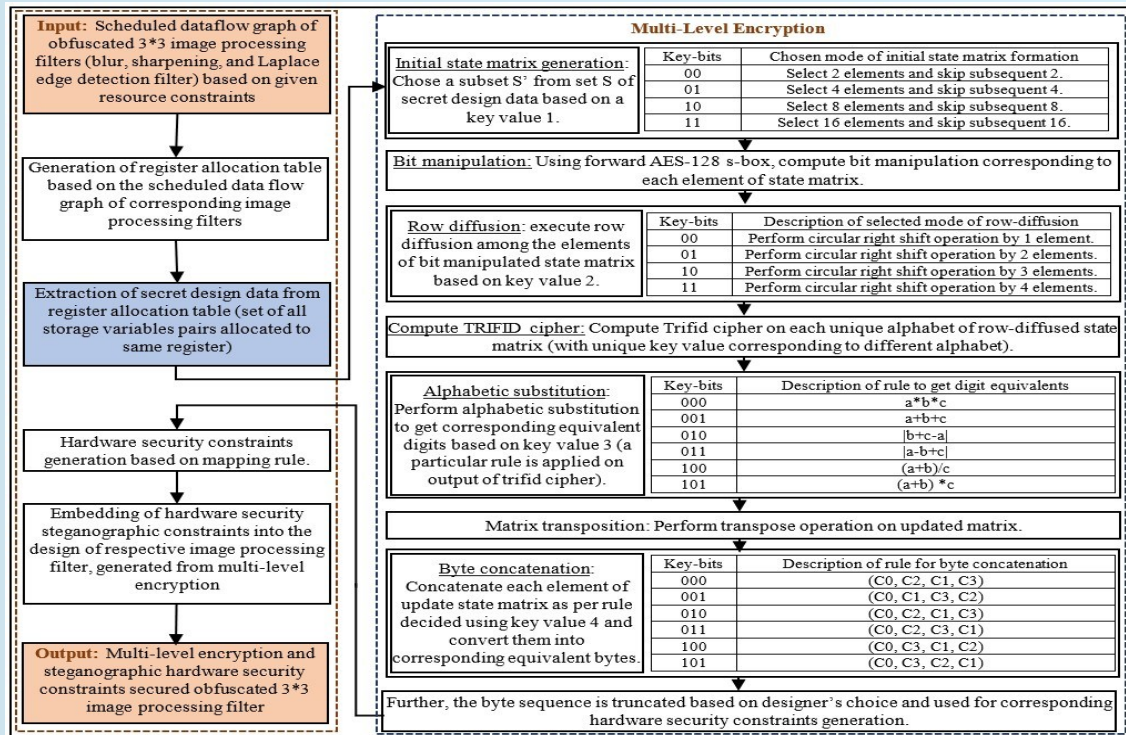


Fig.2. Flow-chart of proposed methodology for securing image processing filters using multi-level encryption and steganographic hardware security constraints.

Proposed Work : Extraction of secret design data from SDFG of image processing application

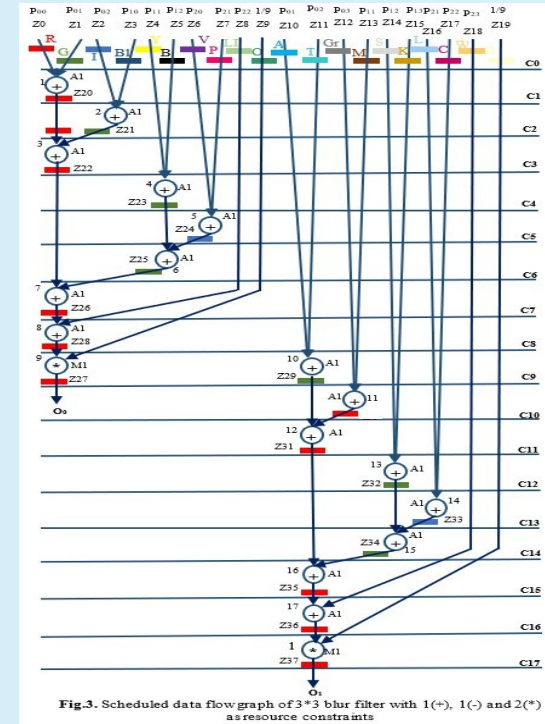


Determination of secret design data based on scheduled data flow graph of image processing application

TABLE I
REGISTER ALLOCATION TABLE OF 3*3 BLUR FILTER DEPICTED IN FIG. 3

CS	Red(R)	Green (G)	Indigo (I)	Blue (BL)	Yellow (Y)	Black (B)	Violet (V)	Pink (P)	Lime (LI)	Olive (O)	Aqua (A)	Teal (T)	Gray (G)	Maroon (M)	Silver (S)	Khaki (K)	Lavender (L)	Crimson (C)	Wheat (W)	Beige (B)
0	Z0	Z1	Z2	Z3	Z4	Z5	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19
1	Z10	Z20	Z2	Z3	Z4	Z5	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19
2	Z20/Z21	Z21/Z20	-	-	Z4	Z5	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19
3	Z22	Z22	-	-	Z4	Z5	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19
4	Z22/Z23	Z23/Z22	-	-	-	-	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19
5	Z22/Z23	Z23/Z22	Z24	Z24	-	-	-	-	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19
6	Z22/Z25	Z25/Z22	-	-	-	-	-	-	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19
7	Z26	Z26	-	-	-	-	-	-	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19
8	Z27	-	-	-	-	-	-	-	-	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19
9	Z28/Z29	Z29/Z28	-	-	-	-	-	-	-	-	-	-	-	Z12	Z13	Z14	Z15	Z16	Z17	Z18
10	Z30/Z29	Z29/Z30	-	-	-	-	-	-	-	-	-	-	-	-	Z14	Z15	Z16	Z17	Z18	Z19
11	Z31	-	-	-	-	-	-	-	-	-	-	-	-	-	Z14	Z15	Z16	Z17	Z18	Z19
12	Z31	Z32	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Z16	Z17	Z18	Z19
13	Z31	Z32	Z33	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Z18	Z19
14	Z31	Z34	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Z18	Z19
15	Z35	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Z18	Z19
16	Z36	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Z19
17	Z37	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

$S = \{(0,20), (0,22), (0,26), (0,27), (0,28), (0,30), (0,31), (0,35), (0,36), (0,37), (20,22), (20,26), (20,27), (20,28), (20,30), (20,31), (20,35), (20,36), (20,37), (22,26), (22,27), (22,28), (22,30), (22,31), (22,35), (22,36), (22,37), (26,27), (26,28), (26,30), (26,31), (26,35), (26,36), (26,37), (27,28), (27,30), (27,31), (27,35), (27,36), (27,37), (28,30), (28,31), (28,35), (28,36), (28,37), (30,31), (30,35), (30,36), (30,37), (31,35), (31,36), (31,37), (34,36), (35,37), (36,37), (1,21), (1,23), (1,25), (1,29), (1,32), (1,34), (21,23), (21,25), (21,29), (21,32), (21,34), (23,25), (23,29), (23,32), (23,34), (25,29), (25,32), (25,34), (29,32), (29,34), (32,34), (2,24), (2,33), (33,24)\}$.



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Proposed Work : Implementation of multi-level encryption

Generation of initial state matrix and implementation of multi-level encryption on generated state matrix



$S = \{(0,5), (0,7), (0,B), (0,C), (0,D), (0,1), (0,5), (0,6), (5,7), (5,B), (5,C), (5,D), (5,1), (5,6), (5,7), (7,B), (7,C), (7,D), (7,1), (7,6), (B,C), (B,D), (B,1), (B,6), (C,D), (C,1), (C,6), (D,1), (D,6), (1,6), (1,8), (1,A), (1,E), (1,2), (1,4), (6,8), (6,A), (6,E), (6,2), (6,4), (8,A), (8,E), (8,2), (8,4), (A,E), (A,2), (A,4), (E,2), (E,4), (2,4), (2,3), (3,9)\}$.

TABLE II GENERATED INITIAL STATE MATRIX				TABLE III STATE MATRIX AFTER BIT MANIPULATION (S-BOX)				TABLE IV STATE MATRIX AFTER ROW DIFFUSION			
05	07	0B	0C	6B	C5	2B	FE	C5	2B	FE	6B
57	5B	5C	5D	5B	39	4A	4C	4C	5B	39	4A
7C	7D	71	76	10	FF	A3	38	10	FF	A3	38
CD	C1	16	D1	BD	78	B4	3E	B4	3E	BD	78
1A	1E	12	14	A2	72	C9	FA	FA	A2	72	C9
64	8A	8E	82	43	7E	19	13	43	7E	19	13
E2	E4	24	23	98	69	36	26	69	36	26	98



Proposed Work : Multi-level encryption

Trifid cipher computation and alphabetic substitution



- ▶ Computing TRIFID cipher on "A":
- ▶ Let IP vendor selected key: EDRFTV\$QAWSZMXNCBGYHUJIKOLP
- ▶ Here, row number (a) is 3, column number (b) is 3, and square matrix (c) number is 1. The state corresponding to "A" is 331. Similarly, the state corresponding to the remaining alphabets is computed based on chosen key.

Square matrix 1			Square matrix 2			Square matrix 3		
E	D	R	W	S	Z	Y	H	U
F	T	V	M	X	N	J	I	K
\$	Q	A	C	B	G	O	L	B

TABLE V
FINAL OBTAINED DIGIT EQUIVALENTS AFTER ALPHABETIC SUBSTITUTION

Assumed key	Alphabet	Corresponding TRIFID cipher state	Defined rule	Output
100	A	331	$(a+b)/c$	6
010	B	122	$ b+c-a $	3
101	C	222	$(a+b) * c$	8
011	D	233	$ a-b+c $	2
000	E	212	$a*b*c$	4
001	F	313	$a+b+c$	7



Proposed Work : Signature generation

Generation of multi-level encryption based signature



TABLE VI
STATE MATRIX AFTER
ALPHABETIC SUBSTITUTION

85	23	74	63
48	53	39	46
10	77	63	38
34	34	32	78
76	62	72	89
43	74	19	13
69	36	26	98

TABLE VII
STATE MATRIX AFTER
PERFORMING TRANSPOSE

85	48	10	34	76	43	69
23	53	77	34	62	74	36
74	39	63	32	72	19	26
63	46	38	78	89	13	98

- ▶ The generated final sequence after byte concatenation is:
"85742363484639531077633834783432766289724319137469 362698".
- ▶ The generated final signature through the proposed approach is:
"10001011111001011110111001000100110111001101
11101111111011111000111001111000111001110111110110101000100111110100111100111111110011
01001111101011010011000".
- ▶ The generated signature is mapped to its corresponding hardware security constraints as per the IP vendor selected mapping rule (if encoding bit of signature is '0' then embed an edge between (even, even) storage variable pair, otherwise embed an edge between (odd, odd) storage variable pair). The generated hardware security constraints are <Z0,Z2>, <Z0,Z4>, <Z0,Z6>,-----,<Z12,Z16>, <Z12,Z18>, <Z1,Z3>, <Z1,Z5>,-----,<Z7,Z19>, <Z7,Z21>.



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Security metrics

Evaluation parameters



➤ Evaluation of Robustness Using Probability of Coincidence (Pc):

$$Pc = \left(1 - \frac{1}{x}\right)^z$$

Where 'x' denotes the number of registers used in the CIG and 'z' denotes the number of hardware constraints added.

➤ Tamper tolerance:

$$TT = q^t$$

Where 'q' and 't' are types of encoding bits present in the mapping rule and strength (size) of generated security constraints respectively.

➤ Design cost:

$$\text{Cost} = t1 * \frac{\text{Area}}{\text{Max area}} + t2 * \frac{\text{Latency}}{\text{Maximum latency}}$$

Where 'area' and 'latency' represents the total area and latency (delay) of the proposed methodology-based secured IP core design; 'max area and max latency' depict the maximum area and latency of the proposed secured design of IP core using maximum resource constraints possible. 't1 and t2' are the weighing factors (weightage given to are and delay), which in the proposed approach is 0.5 each.



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Comparative Analysis

Comparison of Probability of coincidence (P) between proposed and [7] and design cost comparison before and after embedding security constraints



TABLE VIII
Area, Latency, Cost, and Resource configuration of proposed hardware security methodology

Benchmarks	Resource configuration	Baseline design (before signature embedding)			Signature embedded design			Design cost overhead %
		Design area (um)	Design latency (ps)	Design cost	Design area (um)	Design latency (ps)	Design cost	
Blur filter	1(+), 1(*)	110.10	1523.58	0.673	110.10	1523.58	0.673	0
Sharpening filter	1(+), 1(*)	111.67	1921.04	0.675	111.67	1921.04	0.675	0
Laplace edge detection filter	1(+), 1(*)	105.38	1258.61	0.722	105.38	1258.61	0.722	0

TABLE IX
 P_C and TT comparison between proposed and [7]

Benchmarks	P_C (Proposed)	TT (Proposed)	P_C [7]	TT [7]
Blur filter	3.71E-04	2.28E+46	5.98E-01	4.19E+06
Sharpening filter	1012E-03	8.92E+43	5.72E-01	1.04E+06
Laplace edge detection filter	3.87E-04	8.11E+31	5.52E-01	3.27E+04



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